SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I L.Sathyam sharma (1925260282) *(Name / Reg No) certify that this* submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L Sath

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :

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1. A HYPERVISOR SOFTWARE :-

A hypervisor is a software that creates and manages virtual machines by sharing the resources of physical computer

TYPE-1 HYPERVISOR :-

It provides efficient resources allocation strong workload isolation and better control over enterprise infrastructure.

FEATURE :-

ARCHITECTURE :- Runs directly on hardware

PERFORMANCE :- High performance

SECURITY :-
Small host software layer reduce attack exposure

MANAGEABILITY :-
Supports centralized enterprise management

REALIABILITY :-
Suitable for continous enterprise workloads

Example :-
vmware, Microsoft Hyper-V xan

Type-2 Hypervisor :-
These are convienient for development and testing but introduce and additional host OS layer.

Feature :-

Architecture :-
Runs on host OS

Performance :-
comparability lower performance

Security :-
Depends on the security of the host OS and hypervisor

Manageability :-
Easier for the individual users and testing

Reliability :-
more affected by host OS Failures

Examples :-
oracle, virtual Box, VMware.
workstation.

2. cloud providers may run finance, health care and student Portal application on the same physical hardware. virtualization allows there workload to operate independently.

How isolation is activated :-

VM VISULATION :-
Each workload runs in a separate VM with its own operation system and allocated resources

Hyper visor control :-
The hypervisor manages CPU, Memory, storage and hardware access between VMs

Network isolation :-
VLANS, virtual network and Firewall rules separate traffic between departments

Access control :-
Role - based access control (RBAC) restricts users and administrators to authorized resources

storage isolation :-
Separate virtual disks, encryption and access permissons protects sensitive information.

TWO mayor attack surfaces :-

Hypervisor vulnerabilities :-
An attacker exploiting a hypervisor flaw may Escape a VM and gain access to the host

Virtual network and management interfaces:-
weak authentication, misconfigured network rules, or exposed management APIs., may allow unauthorized access

3 1. CPU - utilization - 92%.
* The host is approaching CPU saturation
* Too many VMs or CPU-intensive application may be running
*correction:- Migrate some VMs to other hosts, balance workloads and add CPU capacity if necessary

Memory utilization - 88%.
* Limited free ~~memory~~ memorey may cause memory contention or swapping

* correction:- Reduce unnecessary memory allocation investigate memory leaks and add RAM or redistribute VM

Storage I/O Latency - 35 cm :-

* Slow disk operations may delay applications requests and VM startup
* Possible causes included overloaded storage devices excessive I/O request or storage network congestion

*correction :-

check storage performance matrics reduce unnecessary disk activity and move workloads to faster SSD or VMs storage when appropriate

VM-boat time - 170 sec

* slow booting may results from cpu contention storage delays, memory pressure or excessive startup services

correction :-

optimize VM templates, investigate boot logs, improve storage performance and stagger mass VM startups

* Software defined Datacentre ( )
uses software to manage and automate computing storage and networking resources instead of relaying mainly on manually configured physical infrastructure

Traditional datacentre :-

compute : manual server provisioning

storage :- physical storage configured manually

Network : manual switch and router configuration

Scalability :- often requires manual hardware configuration

Software - Defined data centre

compute - Automated VM Provisioning

storage - Software defined storage pools

Network - Software defined networking and Policies

scalability - Resources can be allocated dynamically

compute virtualization:-

Hypervisor allows multiple VMs to share physical servers. improving utilization and enabling rapid deployment

storage virtualization:-

combines physical storage into logical pools making capacity allocation replication and expansion ea and Expansion easier

Network virtualization:-

creates logical networks, virtual switches and security polices independently of physical network topology

5 cloud Federation allows users and applications to access resources across multiple cloud providers using trusted identity relationships

identity mismatches :-

* Different identity formats and user identifiers
* Incompatible authentication process
* Different roles name and permissions models
* Incorrect token claims . audience values or expiration settings
* Inconsistent user provisoning and deprovisioning

PRoposed secure identity and privacy design :-

1. central identity providers :-

uses a trusted identity provider to manage user identities and authentication centrally.

2. standards - based Federation :-

uses open ID connect ( ) or SAML 2.0 for Federation authentication; depending on the providers and applications supported.

3. Identity mapping :-

map users, groups and roles consistensly across cloud, a Finance analyst should receive only the corresponding Finance

4. Strong authentication :-
Implement multi-factor authentication
( ) short-lived tokens. and secure
sessions management

5 least-privilege access
Orant users only the permissions
required for their jobs use role-
based or attribute-based access control
and separte sensitive workloads

6. Privacy protection :-
Encrypt data in transit and at rest
minimize personal information shared
btwn providers
use appropriate data residency and
retention policies
Audit identity and access events across
all clouds